



```
In [1]: import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import pandas as pd
%matplotlib inline
```

```
In [2]: ## Load the dataset
from sklearn.datasets import load_breast_cancer
```

```
In [3]: cancer_dataset=load_breast_cancer()
```

```
In [4]: cancer_dataset.keys()
```

```
Out[4]: dict_keys(['data', 'target', 'frame', 'target_names', 'DESCR', 'feature_names', 'filename'])
```

```
In [5]: print(cancer_dataset.DESCR)
```

```
.. _breast_cancer_dataset:
```

```
Breast cancer wisconsin (diagnostic) dataset
```

```
-----
```

```
**Data Set Characteristics:**
```

```
:Number of Instances: 569
```

```
:Number of Attributes: 30 numeric, predictive attributes and the class
```

```
:Attribute Information:
```

- radius (mean of distances from center to points on the perimeter)
- texture (standard deviation of gray-scale values)
- perimeter
- area
- smoothness (local variation in radius lengths)
- compactness (perimeter² / area - 1.0)
- concavity (severity of concave portions of the contour)
- concave points (number of concave portions of the contour)
- symmetry
- fractal dimension ("coastline approximation" - 1)

The mean, standard error, and "worst" or largest (mean of the three worst/largest values) of these features were computed for each image, resulting in 30 features. For instance, field 0 is Mean Radius, field 10 is Radius SE, field 20 is Worst Radius.

- class:
 - WDBC-Malignant
 - WDBC-Benign

```
:Summary Statistics:
```

=====	=====	=====
	Min	Max
=====	=====	=====
radius (mean):	6.981	28.11
texture (mean):	9.71	39.28
perimeter (mean):	43.79	188.5
area (mean):	143.5	2501.0
smoothness (mean):	0.053	0.163
compactness (mean):	0.019	0.345
concavity (mean):	0.0	0.427
concave points (mean):	0.0	0.201
symmetry (mean):	0.106	0.304
fractal dimension (mean):	0.05	0.097
radius (standard error):	0.112	2.873
texture (standard error):	0.36	4.885
perimeter (standard error):	0.757	21.98
area (standard error):	6.802	542.2
smoothness (standard error):	0.002	0.031
compactness (standard error):	0.002	0.135
concavity (standard error):	0.0	0.396

concave points (standard error):	0.0	0.053
symmetry (standard error):	0.008	0.079
fractal dimension (standard error):	0.001	0.03
radius (worst):	7.93	36.04
texture (worst):	12.02	49.54
perimeter (worst):	50.41	251.2
area (worst):	185.2	4254.0
smoothness (worst):	0.071	0.223
compactness (worst):	0.027	1.058
concavity (worst):	0.0	1.252
concave points (worst):	0.0	0.291
symmetry (worst):	0.156	0.664
fractal dimension (worst):	0.055	0.208
=====	=====	=====

:Missing Attribute Values: None

:Class Distribution: 212 - Malignant, 357 - Benign

:Creator: Dr. William H. Wolberg, W. Nick Street, Olvi L. Mangasarian

:Donor: Nick Street

:Date: November, 1995

This is a copy of UCI ML Breast Cancer Wisconsin (Diagnostic) datasets.
<https://goo.gl/U2Uwz2>

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image.

Separating plane described above was obtained using Multisurface Method-Tree (MSM-T) [K. P. Bennett, "Decision Tree Construction Via Linear Programming." Proceedings of the 4th Midwest Artificial Intelligence and Cognitive Science Society, pp. 97-101, 1992], a classification method which uses linear programming to construct a decision tree. Relevant features were selected using an exhaustive search in the space of 1-4 features and 1-3 separating planes.

The actual linear program used to obtain the separating plane in the 3-dimensional space is that described in:
 [K. P. Bennett and O. L. Mangasarian: "Robust Linear Programming Discrimination of Two Linearly Inseparable Sets", Optimization Methods and Software 1, 1992, 23-34].

This database is also available through the UW CS ftp server:

ftp ftp.cs.wisc.edu
 cd math-prog/cpo-dataset/machine-learn/WDBC/

.. topic:: References

- W.N. Street, W.H. Wolberg and O.L. Mangasarian. Nuclear feature extraction for breast tumor diagnosis. IS&T/SPIE 1993 International Symposium on Electronic Imaging: Science and Technology, volume 1905, pages 861-870, San Jose, CA, 1993.
- O.L. Mangasarian, W.N. Street and W.H. Wolberg. Breast cancer diagnosis and prognosis via linear programming. Operations Research, 43(4), pages 570-577, July-August 1995.
- W.H. Wolberg, W.N. Street, and O.L. Mangasarian. Machine learning techniques to diagnose breast cancer from fine-needle aspirates. Cancer Letters 77 (1994) 163-171.

```
In [6]: df=pd.DataFrame(cancer_dataset['data'],columns=cancer_dataset['feature_names']
df.head()
```

```
Out[6]:
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	mean cor p
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0.1
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0.0
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0.1
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0.1
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0.1

5 rows × 30 columns

```
In [7]: ## Standardization
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
```

```
In [8]: scaler.fit(df)
```

```
Out[8]: StandardScaler()
```

```
In [9]: scaled_data=scaler.transform(df)
```

```
In [10]: scaled_data
```

```
Out[10]: array([[ 1.09706398, -2.07333501,  1.26993369, ...,  2.29607613,
                  2.75062224,  1.93701461],
                [ 1.82982061, -0.35363241,  1.68595471, ...,  1.0870843 ,
                  -0.24388967,  0.28118999],
                [ 1.57988811,  0.45618695,  1.56650313, ...,  1.95500035,
                  1.152255   ,  0.20139121],
                ...,
                [ 0.70228425,  2.0455738 ,  0.67267578, ...,  0.41406869,
                  -1.10454895, -0.31840916],
                [ 1.83834103,  2.33645719,  1.98252415, ...,  2.28998549,
                  1.91908301,  2.21963528],
                [-1.80840125,  1.22179204, -1.81438851, ..., -1.74506282,
                  -0.04813821, -0.75120669]])
```

```
In [11]: ## Applying PCA Algorithms
        from sklearn.decomposition import PCA
```

```
In [29]: pca=PCA(n_components=2)
```

```
In [30]: data_pca=pca.fit_transform(scaled_data)
```

```
In [31]: data_pca
```

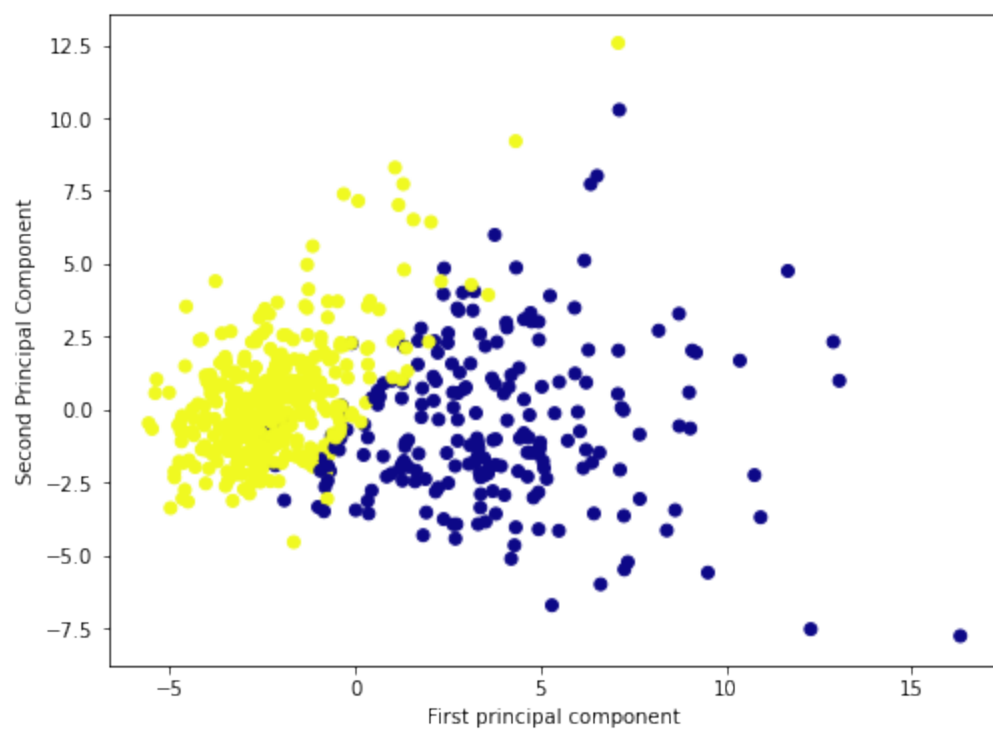
```
Out[31]: array([[ 9.19283683,  1.94858307],
                [ 2.3878018 , -3.76817174],
                [ 5.73389628, -1.0751738 ],
                ...,
                [ 1.25617928, -1.90229671],
                [10.37479406,  1.67201011],
                [-5.4752433 , -0.67063679]])
```

```
In [32]: pca.explained_variance_
```

```
Out[32]: array([13.30499079,  5.7013746 ])
```

```
In [33]: plt.figure(figsize=(8,6))
        plt.scatter(data_pca[:,0],data_pca[:,1],c=cancer_dataset['target'],cmap='plasma')
        plt.xlabel('First principal component')
        plt.ylabel('Second Principal Component')
```

```
Out[33]: Text(0, 0.5, 'Second Principal Component')
```



In []: